

PRODUCT SALES PERFORMANCE

2023 - 2025

A. OBJECTIVE

The objective of this project is to develop a dynamic Power BI dashboard that provides a comprehensive analysis of product sales performance for the years 2023 - 2025. The dashboard will enable stakeholders to monitor key performance indicators, identify sales trends, understand customer behavior, and generate actionable insights to support data-driven business decisions.

B. PROBLEM STATEMENT

The current sales data contains information related to customer purchases, product performance, revenue, quantity sold, payment methods, and customer ratings. A centralized dashboard is required to consolidate this information and provide stakeholders with an interactive view of sales performance and customer purchasing behavior.

C. GOALS

- ✓ Monitor overall sales and revenue performance.
- ✓ Identify high- and low-performing products.
- ✓ Understand customer purchasing behavior.
- ✓ Identify sales trends across months, quarters, and years.
- ✓ Analyze customer preferences through payment methods and ratings.

D. USERS

Stakeholder	Purpose
Sales Manager	Monitor sales and product performance
Business Manager	Review overall business performance
Product Manager	Identify top and underperforming products
Business Analyst	Analyze trends and customer behavior
Management	Review KPIs and support decision-making

E. SCOPE

In Scope	Out of Scope
Product sales analysis	Inventory management
Revenue analysis	Supplier/vendor analysis
Quantity sold	Employee performance
Payment method analysis	Marketing campaign analysis
Product performance	Product cost/profitability (<i>unless cost data is available</i>)
Monthly/quarterly/yearly trends	--

F. FUNCTIONAL REQUIREMENTS

FR-01: The dashboard shall display KPI's and sales data for the selected date range and product category filters.

FR-02: Users shall be able to apply filters and slicers to dynamically update the KPIs and visualizations based on the selected criteria.

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FR-04: The dashboard shall display quick highlights based on data displayed on the respective dashboard page.

FR-06: The dashboard shall provide button-based navigation between different report pages, allowing users to move between the Sales Overview and Best/Worst Products sections.

FR-06: The dashboard shall display KPI's in 4 cards on both the dashboards.

BUSINESS RULES (KPI CARDS)

✓ **Total Number of Products Purchased:**

The system shall calculate the total number of product purchases by counting every product purchase record in the *CustomerPurchaseHistory* table. Each product is counted individually, even when the same customer purchases multiple products.

✓ **Total revenue:**

The system shall calculate the total revenue generated from all customer purchases by adding the Total Price value for every purchase transaction recorded in the *CustomerPurchaseHistory* table.

✓ **Total quantity sold:**

The system shall calculate the total quantity of products sold by adding the Quantity value from every purchase transaction recorded in the *CustomerPurchaseHistory* table.

✓ **Average order value:**

The system shall calculate the Average Order Value (AOV) by dividing the total revenue generated from customer purchases by the number of distinct products purchased.

BUSINESS RULES (HOME DASHBOARD)

➤ **Total quantity of products sold quarterly and yearly:**

The system shall calculate the total quantity of products sold by year and quarter based on PurchaseDate. The results shall be displayed in a Clustered Bar Chart for easy comparison of sales across periods.

➤ **Total quantity sold per month:**

The system shall calculate the total quantity of products sold by month based on PurchaseDate. The results shall be displayed in a Clustered Bar Chart to compare monthly product sales.

➤ **Product-category-wise Customer Count:**

The system shall calculate the total number of customers by product category. The results shall be displayed in a Tree Map to compare customer distribution across product categories.

➤ **Sales performance for the last three years:**

The system shall calculate the total revenue generated by each product. The results shall be displayed in a Line Chart to compare revenue across products.

➤ **% of payment methods used:**

The system shall calculate the percentage of payments made through each payment method. The results shall be displayed in a Donut Chart to show the proportion of each payment method.

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BUSINESS RULES (BEST / WORST DASHBOARD)

➤ Top 3 products by Revenue:

The system should identify the top 3 products by total revenue. The results shall be displayed in a Bar Chart to compare revenue performance among the top-performing products.

➤ Bottom 3 products by Revenue:

The system should identify the bottom 3 products by total revenue. The results shall be displayed in a Bar Chart to compare revenue performance among the top-performing products

➤ Top 3 products by Quantity:

The system shall identify the top 3 products with the highest quantity sold. The results shall be displayed in a Bar Chart to compare product sales volume.

➤ Bottom 3 products by Quantity:

The system shall identify the bottom 3 products with the highest quantity sold. The results shall be displayed in a Bar Chart to compare product sales volume.

➤ Top 3 products by Rating:

The system shall identify the top 3 products with the highest number of 5-star ratings. The results shall be displayed in a Bar Chart to compare highly rated products.

➤ Bottom 3 products by Rating:

The system shall identify the bottom 3 products with the lowest number of 1-star ratings. The results shall be displayed in a Bar Chart to compare highly rated products.

ACCEPTANCE CRITERIA

Dashboard will be considered complete when:

- ✓ All defined KPIs are displayed correctly.
- ✓ Revenue and quantity calculations match the source data.
- ✓ Filters dynamically update all relevant visuals.
- ✓ Year/quarter/month analysis displays the correct time period.
- ✓ Top and Bottom 3 products are calculated correctly.
- ✓ Navigation buttons work between dashboard pages.
- ✓ Visuals are clearly labelled and easy to interpret.
- ✓ Data is presented consistently across all report pages.